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Subject - Quant

Assignment - 02

Q1. Find the perimeter of an equilateral triangle whose side is 8 cm.

Solⁿ:- Side = 8 cm

Perimeter of an equilateral triangle

$$= 3 \times \text{Side}$$

$$= 3 \times 8$$

$$\boxed{P = 24 \text{ cm}}$$

Q2. Find the area of an equilateral triangle whose side is 10 cm.

Ans → Side = 10 cm

Area of an equilateral $\Delta = \frac{\sqrt{3}}{4} \times a^2$

$$= \frac{\sqrt{3}}{4} \times 10 \times 10$$

$$= \frac{\sqrt{3}}{4} \times 100$$

$$= \sqrt{3} \times 25$$

$$\boxed{\text{Area} = 25\sqrt{3} \text{ cm}^2}$$

Q3. Find the height of an equilateral triangle whose side is 12 cm.

Solⁿ:- Side = 12 cm

$$\text{height} = \frac{\sqrt{3}}{2} a$$

$$= \frac{\sqrt{3}}{2} \times 12$$

$$\boxed{h = 6\sqrt{3} \text{ cm}}$$

Q4. The area of an equilateral triangle is $25\sqrt{3} \text{ cm}^2$, find the length of its side.

Soln:- Area = $25\sqrt{3} \text{ cm}^2$.

$$\frac{\sqrt{3}}{4} a^2 = 25\sqrt{3}$$

$$a^2 = \frac{25}{3} \times \frac{4}{\sqrt{3}}$$

$$a^2 = \frac{100}{3\sqrt{3}}$$

$$a = \sqrt{\frac{100}{3\sqrt{3}}} \text{ cm}$$

Q5. If the perimeter of an equilateral triangle is 36 cm, find its area.

Soln:- Perimeter = 36 cm

$$\text{side} \times 3 = 36$$

$$\text{side} = \frac{36}{3}$$

$$\boxed{\text{side} = 12 \text{ cm}}$$

$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 12 \times 12$$

$$\boxed{A = 36\sqrt{3} \text{ cm}^2}$$

Q6. Find the area of a rectangle whose length = 15 cm & breadth = 8 cm.

Ans. → length = 15 cm

b = 8 cm

$$\text{Area of rectangle} = l \times b$$

$$= 15 \times 8$$

$$\boxed{\text{Area} = 120 \text{ cm}^2}$$

Q7. Find the perimeter of a rectangle whose length = 12 cm + breadth = 5 cm

Soln: Perimeter = $2(l+b)$
 $= 2(12+5)$

$$= 2 \times 17$$
$$\boxed{P = 34 \text{ cm}}$$

Q8. The area of a rectangle is 96 cm^2 and its breadth is 8 cm. Find its length.

Soln: - Area = 96 cm^2 , $b = 8 \text{ cm}$

$$l \times b = 96 \text{ cm}^2$$

$$l \times 8 = 96 \text{ cm}^2$$

$$l = \frac{96 \text{ cm}^2}{8}$$

$$\boxed{l = 12 \text{ cm}}$$

Q9. Find the area of a parallelogram whose base is 14 cm + height = 9 cm.

Soln: - base = 14 cm
height = 9 cm

$$\text{Area} = B \times H$$

$$= 14 \times 9$$

$$\boxed{\text{Area} = 126 \text{ cm}^2}$$

Q10. The base of a parallelogram is 18 cm and the height is 7 cm. Find its area.

Soln: - Area = 18×7

$$\boxed{A = 126 \text{ cm}^2}$$